

policies. Acknowledging its importance during the pandemic, the government has awarded infrastructure status to data centres, with even state governments offering subsidised land, power subsidies, exemptions on stamp duty, discounts on usage of renewable energy and procurement of IT components made locally, etc. The government's initiative to build a backbone for digital transactions and authorisation with the JAM trinity (Jan Dhan-Aadhaar-Mobile), UPI, and India Stack APIs, proved to be a major catalyst in driving the explosion of data generation. And this could only be scaled on the backbone of domestic data centres.

Then in July 2018, the Data Protection Act, prepared by the Srikrishna Committee, suggested that citizens'

ommends data transfer to friendly countries subject to permissions, the law is unclear on how easy or cumbersome this process could be. But industry experts believe this is unlikely to impact the data centre boom.

With the government's digital initiative and support from various stakeholders, a significant transformation has been witnessed in sectors such as education, healthcare, e-commerce and life sciences. "This has led to strong growth for data centre storage and computing, which is reflected in the continued growth of the Indian data centre industry," says Seema Ambastha, CEO of BAM Digital Realty, who expects 681 MW data centre capacity to be added by end-2024, which will double India's data centre capacity to 1,318 MW. BAM

BUILDING A MODERN DATA CENTRE



INFRASTRUCTURE

Designing and building a data centre takes anywhere between 12 and 24 months and they should be able to withstand natural calamities



POWER

Data centres require a lot of uninterrupted high-quality power and typically have back-up generators and other redundant power systems



COOLING

High temperatures and humidity are harmful for these systems. High intensity cooling is required to prevent them from excessive heating



MONITORING/ MANAGEMENT

They use monitoring and management tools to track performance, identify issues and make adjustments as needed



SERVERS

These are the primary computing devices. They store and process data and run the software applications that power various operations

data could be protected by storing it locally. This move led to many companies ramping up data centre capacity. "This led to increased interest in India with local players ramping up new developments aggressively and foreign players looking to enter India and capture a part of the pie," says Devi Shankar, President-Industrial & Logistics and Data Center, ANAROCK Capital. Amazon invested around \$197 million in its data services arm in the country. Not to be left behind, companies like Google and Microsoft also announced similar plans. Even markets regulator Sebi announced its intention to develop guidelines that would mandate foreign entities to store data pertaining to India locally. It was a good enough trigger for the data centre industry.

Although the new Personal Data Protection Act rec-

Digital Realty—a JV between Digital Realty and Brookfield Infrastructure—plans to come up with its first data centre in India by December 2023 in Chennai, with more than 100 MW capacity. Globally, Digital Realty has over 310 data centres and the JV plans to invest \$2 billion over the next few years, say industry sources.

According to the Ministry of Electronics & IT, the size of the digital economy in India is estimated to grow from \$200 billion in 2017-18 to \$1 trillion by 2025, "which further necessitates strong growth of data centres", says Ranganath Sadasiva, CTO of Hybrid IT at Hewlett Packard Enterprise, India. Due to this explosion of data emanating from the growth of the digital economy, some enterprises are also adopting cloud-only models. And India is fast emerging as the world's